

Comparative Overview of Agentic Orchestration Frameworks for AWS and Databricks

1. Introduction: Defining Agentic Orchestration Frameworks in Cloud Environments

Agentic artificial intelligence (AI) systems represent a paradigm shift in automation, leveraging the advanced reasoning capabilities of Large Language Models (LLMs) to autonomously determine actions, execute tasks, and integrate with external tools ¹. These sophisticated systems are designed to tackle complex workflows and enhance overall productivity by significantly reducing the need for human intervention ¹. The underlying mechanisms that enable the development, management, and scaling of these autonomous systems are provided by agentic orchestration frameworks. The increasing sophistication and accessibility of LLMs have acted as a significant catalyst in the evolution and widespread adoption of agentic AI frameworks ². This close relationship suggests that the capabilities of these orchestration tools are intrinsically linked to the advancements in LLM technology, with improvements in LLMs directly influencing the potential and complexity of the frameworks themselves.

The application of agentic AI is expanding across a diverse range of industries, including sectors such as automated trading, healthcare, manufacturing, finance, supply chain management, and customer support ². This broad applicability indicates a growing recognition of the versatility and adaptability of agentic frameworks in addressing various business needs and problem domains. Consequently, the specific demands placed on these frameworks, particularly in terms of scalability, integration, and monitoring, can vary considerably depending on the specific use case and industry.

Cloud computing environments, such as Amazon Web Services (AWS) and Databricks, offer the robust and scalable infrastructure, along with a comprehensive suite of services, that agentic systems can effectively utilize. Orchestration frameworks play a crucial role in managing the intricate interactions between multiple agents, LLMs, external tools, and the various cloud services to achieve complex, overarching objectives. Therefore, the selection of an appropriate framework should be carefully aligned with the specific capabilities and services offered by both AWS and Databricks to ensure optimal efficiency and seamless integration. The user's explicit mention of both AWS and Databricks highlights a specific need for frameworks that can operate effectively and integrate smoothly with the services and data residing within these distinct platforms. This requirement for cross-platform compatibility and integration is a key consideration in the evaluation process. Organizations that utilize both AWS and Databricks often have data and services distributed across these environments, making it essential for an agentic framework to effectively bridge this gap and facilitate the efficient access and utilization of these distributed resources.

2. Top Agentic Orchestration Frameworks for AWS and Databricks

Based on current trends and adoption rates, several agentic orchestration frameworks have emerged as popular choices for building sophisticated AI systems². Among the leading contenders are LangChain (along with its extension LangGraph), CrewAI, Microsoft Semantic Kernel, Microsoft AutoGen, the AWS Multi-Agent Orchestrator (specifically tailored for AWS), and the Databricks Mosaic AI Agent Framework (designed for the Databricks environment). It is important to note that while frameworks like LangChain, LangGraph, CrewAI, Semantic Kernel, and AutoGen are often presented as generally applicable, the AWS Multi-Agent Orchestrator and Databricks Mosaic AI Agent Framework exhibit a closer coupling with their respective cloud platforms². This suggests a potential trade-off where platform-specific frameworks might offer enhanced optimization and access to native services but could potentially limit flexibility in scenarios requiring multi-cloud deployments.

2.1. LangChain

LangChain is recognized as a versatile, open-source framework specifically engineered to simplify the process of developing applications powered by large language models³. It excels in critical areas such as managing conversational context, maintaining memory across interactions, and integrating with a wide array of external tools³. The framework's modular design enables developers to seamlessly connect LLMs with various resources, including APIs, databases, and memory systems, allowing for the creation of context-aware responses³. Backed by a robust and active community, along with comprehensive documentation, LangChain is accessible to both novice and experienced developers³. It has found widespread application in building conversational agents, developing retrieval-augmented systems, and automating complex workflows².

LangChain offers extensive integration capabilities with the AWS ecosystem, including services like Amazon Bedrock for accessing a variety of foundation models, Amazon Kendra for intelligent search, Amazon SageMaker for machine learning services, Amazon S3 for object storage, Amazon Textract for document analysis, Amazon Athena for serverless interactive queries, and AWS Glue for data integration⁷. Similarly, LangChain provides robust integration with the Databricks platform, allowing connection with Databricks SQL for data processing, leveraging HuggingFace models deployed on Databricks, utilizing Databricks Open Models, and integrating with models served through Databricks Model Serving⁹. The breadth of these integrations positions LangChain as a strong contender for organizations seeking a cross-platform agentic orchestration solution. Its inherent modularity and the strong support from its community further enhance its appeal for diverse use cases. The sheer number of available integrations underscores LangChain's commitment to interoperability and its ability to effectively leverage the diverse range of services offered by both AWS and Databricks.

2.2. LangGraph

Built upon the foundation of LangChain, LangGraph extends its capabilities to facilitate the development of stateful, multi-agent systems³. It introduces a graph-based approach where workflows are represented as interconnected nodes and edges, providing developers with

precise control over intricate processes and interactions between agents ³. This framework is particularly well-suited for applications that demand advanced memory management, robust error recovery mechanisms, and the incorporation of human-in-the-loop features ³. LangGraph supports the implementation of cyclical graphs and branching logic, enabling more dynamic control over the behavior of agents ⁵.

In terms of cloud integration, LangGraph seamlessly integrates with AWS Bedrock, allowing for the construction and deployment of sophisticated multi-agent systems within the AWS cloud environment ⁸. Furthermore, LangGraph is a supported authoring framework within the Databricks Mosaic AI Agent Framework, enabling developers to leverage its graph-based approach to create and manage agents on the Databricks platform ¹³. LangGraph's graph-based architecture offers a more structured and intuitive way to manage complex interactions among multiple agents compared to the linear chain-of-thought approach often employed in LangChain. Its availability and support on both AWS and Databricks make it a relevant and powerful option for users seeking to orchestrate intricate agentic workflows across these platforms. The graph structure inherent in LangGraph facilitates the creation of more sophisticated control flows and the management of state in complex multi-agent scenarios, which can be essential for applications such as advanced decision-making systems or intricate simulations.

2.3. CrewAI

CrewAI stands out as an intuitive framework specifically designed for multi-agent collaboration, aiming to replicate the dynamics of human teams ². It simplifies the process of creating role-based AI agents that can work together cohesively on a variety of tasks, requiring minimal coding and offering an easy setup process ³. CrewAI is particularly well-suited for rapid prototyping and excels in scenarios where seamless coordination between agents is crucial, such as in logistics or resource planning ². Built upon LangChain, CrewAI benefits from its extensive ecosystem of tools ³.

Regarding cloud integration, CrewAI can integrate with various AWS services through its use of LiteLLM, a library that provides a unified interface for connecting to a wide range of LLM providers, including AWS Bedrock and SageMaker ¹⁶. Similarly, CrewAI can establish connections with LLMs hosted on Databricks Model Serving via LiteLLM ¹⁷. Additionally, CrewAI offers integration with Databricks Unity Catalog, allowing for the efficient management and utilization of tools within the Databricks environment ¹⁹. CrewAI's primary focus on facilitating team-like collaboration among agents, coupled with its relatively straightforward setup compared to frameworks like LangGraph, can be particularly appealing for use cases where modeling human team dynamics is advantageous. Its reliance on LiteLLM provides a significant degree of flexibility in connecting with diverse LLM providers available on both AWS and Databricks. The "crew" concept simplifies the mental model required for designing multi-agent systems, making it easier to orchestrate tasks that require collaborative effort.

2.4. Microsoft Semantic Kernel

Microsoft Semantic Kernel is a framework that focuses on integrating AI capabilities into existing enterprise applications, with a strong emphasis on semantic reasoning and context awareness ². It achieves this by combining the power of LLMs with traditional programming paradigms,

offering pre-built connectors that enable seamless integration with various business systems³. Designed to be used with both .NET and Python programming languages, Semantic Kernel is both lightweight and powerful, capable of enhancing decision-making processes in areas such as customer service and IT operations³. Its key strengths lie in its reusable components and ability to retain memory across interactions, making it particularly suitable for building virtual assistants³.

In terms of cloud integration, Semantic Kernel has established direct integration with AWS Bedrock Agents and AWS Bedrock itself, allowing users to access a wide selection of foundation models²⁰. While specific details regarding direct integration with Databricks are less prominent in the provided material, Semantic Kernel's support for the Python language suggests that integration is possible, although it might necessitate more manual configuration compared to frameworks like LangChain. Semantic Kernel's focus on enterprise-level integration and its emphasis on semantic understanding could be particularly beneficial for organizations that have a significant investment in the Microsoft technology stack or those that prioritize semantic reasoning within their AI applications. Its growing support for the AWS platform is also a noteworthy development. The ability to seamlessly blend AI functionalities with established programming practices can make it easier for developers within enterprise environments to adopt agentic capabilities and integrate them into their existing applications.

2.5. Microsoft AutoGen

AutoGen, developed by Microsoft, is an enterprise-grade framework specifically designed for building multi-agent systems, with a strong focus on automation and scalability². It offers comprehensive support for code generation, execution, and seamless collaboration between agents, incorporating robust error handling and logging mechanisms³. AutoGen features an asynchronous, event-driven architecture, which contributes to its ability to handle large-scale operations effectively⁴. Furthermore, the framework includes AutoGen Studio, a no-code interface that broadens its accessibility, while its inherent flexibility caters to the needs of advanced users³. AutoGen is particularly well-suited for production environments that demand high reliability, although its initial setup might be more complex compared to simpler frameworks like CrewAI.

Regarding cloud integration, AutoGen supports integration with AWS Bedrock, enabling users to run inference using a variety of available models²³. In the context of Databricks, AutoGen can be integrated through the Foundation Model APIs, specifically DBRX, and it also allows the use of functions defined in Databricks Unity Catalog as tools for agents²⁴. AutoGen's strong emphasis on automation, its inherent scalability, and its asynchronous architecture make it a compelling choice for organizations dealing with complex, distributed workflows, such as those involved in cloud automation. The availability of a no-code interface through AutoGen Studio can make the framework more accessible to a wider range of users, including those with less extensive coding experience. The focus on asynchronous messaging and event-driven interactions allows for the development of highly scalable and responsive multi-agent systems.

2.6. AWS Multi-Agent Orchestrator

The AWS Multi-Agent Orchestrator is a flexible framework introduced by Amazon Web Services

to specifically manage multiple AI agents and handle complex conversational scenarios within the AWS ecosystem ²⁷. It is designed to intelligently route user queries to the most appropriate agent based on the context of the conversation ²⁷. The framework supports development in both Python and TypeScript, offering flexibility to developers ²⁷. Key features include robust context management across multiple turns of conversation, an extensible architecture that allows for easy integration of new agents, and the ability to deploy universally across AWS Lambda, local environments, and other cloud platforms ²⁷. It also comes equipped with pre-built agents and classifiers to accelerate the development process ²⁸.

The AWS Multi-Agent Orchestrator exhibits deep integration with various AWS services, most notably Amazon Bedrock and AWS Lambda ²⁷. However, the provided information does not indicate any direct integration with the Databricks platform, suggesting that its primary focus is on the AWS ecosystem. This framework is a strong option for organizations that are heavily invested in the AWS cloud, offering specialized features and tight integration with native AWS services. However, its lack of direct integration with Databricks might limit its suitability for users specifically seeking a framework that seamlessly operates across both environments. Its AWS-centric design enables organizations to fully leverage the power and capabilities of the AWS platform for building and deploying sophisticated agentic systems.

2.7. Databricks Mosaic AI Agent Framework

The Databricks Mosaic AI Agent Framework is a comprehensive suite of tools provided by Databricks, specifically designed to assist developers in building, deploying, and evaluating production-quality AI agents ³⁴. It places a strong emphasis on ensuring high production quality, enabling rapid development iterations, and providing robust governance and guardrails for AI applications ³⁴. The framework is seamlessly integrated with the Databricks Data Intelligence Platform, leveraging key services such as Mosaic AI Vector Search for efficient data retrieval, Mosaic AI Model Serving for deploying agents, and Mosaic AI Gateway for managing and monitoring AI usage ³⁴. It supports the use of various agent authoring frameworks, including LangGraph, PyFunc, and OpenAI, offering flexibility in how agents are constructed ¹³. The Mosaic AI Agent Framework also provides built-in features for agent evaluation, tracing agent behavior, and monitoring performance directly within the Databricks environment ³⁴.

While its primary focus is on the Databricks platform, the Mosaic AI Agent Framework does offer integration with AWS services, particularly Amazon Bedrock, allowing users to access foundation models and build Retrieval Augmented Generation (RAG) systems ⁴¹. However, its core strengths and deepest integrations lie within the Databricks ecosystem. For organizations that heavily utilize Databricks for their data processing and AI initiatives, the Mosaic AI Agent Framework is a natural and compelling choice. Its tight integration with the entire Databricks platform provides a streamlined and well-governed experience for developing and deploying agentic applications. By leveraging the full range of Databricks services for data management, model deployment, and governance, organizations can establish a cohesive and efficient platform for building and managing sophisticated agentic applications.

3. Comparative Analysis of Key Features

3.1. Scalability

When considering the scalability of agentic orchestration frameworks, several factors come into play, including the ability to handle increasing workloads, manage complex multi-agent interactions, and efficiently utilize resources. LangChain, with its modular design, allows for scaling individual components as needed. LangGraph's graph-based structure is particularly well-suited for managing intricate and large-scale multi-agent systems³. The representation of agent interactions as a graph in LangGraph offers a more adaptable and potentially more scalable approach for highly complex workflows compared to the linear structure of LangChain. This graph structure can better accommodate parallel processing and dynamic interactions between agents, which are crucial for achieving scalability in multi-agent systems.

CrewAI is designed with multi-agent collaboration as a central tenet, suggesting an inherent ability to scale for tasks that involve teamwork among agents². It also offers the flexibility of being deployed on both cloud and self-hosted infrastructure⁵. Microsoft Semantic Kernel, while emphasizing enterprise integration, implies scalability within that context. Its underlying asynchronous nature also supports handling a greater number of concurrent operations²². Microsoft AutoGen stands out with its explicit focus on automation and scalability for enterprise-grade multi-agent systems, employing an asynchronous, event-driven architecture². This architectural choice, with its emphasis on asynchronous messaging and event-driven interactions, indicates that AutoGen is specifically designed to handle large-scale, distributed agent networks efficiently. Asynchronous communication allows agents to operate independently and concurrently, which can significantly improve overall system throughput and responsiveness when dealing with heavy loads.

The AWS Multi-Agent Orchestrator is designed to manage complex conversational scenarios and can be deployed across various environments, including AWS Lambda, local setups, and other cloud platforms, suggesting scalability within the AWS ecosystem²⁷. Finally, the Databricks Mosaic AI Agent Framework benefits from its deep integration with the scalable Databricks Data Intelligence Platform, leveraging services such as Model Serving and Vector Search to effectively handle large-scale operations and data volumes³⁴.

3.2. Ease of Use

The ease of use of an agentic orchestration framework is a critical factor for developers, influencing the speed of development and the learning curve associated with adoption. LangChain, while highly versatile, can sometimes require careful tuning to achieve production-level stability³. However, its extensive documentation makes it relatively accessible to developers with varying levels of experience. LangGraph, being built on LangChain, inherits some of its characteristics but introduces the added complexity of a graph-based paradigm, which can present a steeper learning curve³. CrewAI distinguishes itself by its intuitive design, focusing on simplicity and enabling rapid prototyping with minimal coding². Its opinionated approach, however, might limit the degree of customization available for more advanced or specialized use cases. Microsoft Semantic Kernel aims to be lightweight yet powerful, allowing developers to integrate AI functionalities into existing applications without requiring significant architectural overhauls². Microsoft AutoGen offers a dual approach to ease of use: AutoGen Studio provides a no-code interface that broadens accessibility to users with less coding expertise, while its underlying flexibility caters to the needs of more advanced developers.

However, the initial setup process for AutoGen can be more involved compared to some of the simpler frameworks ³.

The AWS Multi-Agent Orchestrator is designed with ease of use in mind, offering pre-built components that facilitate quick development and deployment ²⁷. Its universal deployment capability further simplifies the setup process across different environments. Similarly, the Databricks Mosaic AI Agent Framework focuses on streamlining the development process through features like the Agent Evaluation Review App and tight integration with the Databricks user interface ³⁴. The AI Playground within the Mosaic AI Agent Framework also offers a low-code approach to prototyping agents, making it easier for users to get started ⁴⁰.

3.3. Integration Capabilities

The ability of an agentic orchestration framework to integrate with various other services, tools, and data sources is paramount for building comprehensive and effective AI applications. LangChain and LangGraph offer the most extensive range of integrations, supporting a vast array of LLMs, APIs, databases, memory systems, and cloud services on both AWS and Databricks platforms ³. CrewAI, being built on LangChain, naturally inherits its broad tool ecosystem. Additionally, CrewAI integrates with various LLMs through the LiteLLM library and with Databricks Unity Catalog for managing tools ³. Microsoft Semantic Kernel provides pre-built connectors that enable seamless integration with existing business systems and has demonstrated growing integration capabilities with AWS Bedrock ³. Microsoft AutoGen supports integration with external APIs and tools, with specific integrations available for AWS Bedrock and Databricks Foundation Model APIs and Unity Catalog ⁴.

The AWS Multi-Agent Orchestrator is deeply integrated with the AWS ecosystem, supporting services like Bedrock, Lambda, Lex, and Connect ²⁷. The Databricks Mosaic AI Agent Framework is seamlessly integrated with the Databricks Data Intelligence Platform, encompassing Vector Search, Model Serving, AI Gateway, and Unity Catalog. It also offers integration with AWS Bedrock for accessing foundation models ³⁴.

3.4. Monitoring

Effective monitoring is crucial for ensuring the reliability, performance, and quality of agentic AI systems. LangChain provides the LangSmith platform, which allows for comprehensive debugging, testing, and performance monitoring of LangChain applications ⁶. When used within the Databricks environment, LangChain also benefits from integration with MLflow, which offers experiment tracking and observability features ¹¹. LangGraph, being an extension of LangChain, naturally leverages LangSmith for its monitoring needs. CrewAI's performance monitoring capabilities are designed to work in conjunction with Amazon CloudWatch when deployed on the AWS platform ¹⁶. Its integration with Databricks includes tracing functionalities through MLflow ⁴⁷. Microsoft Semantic Kernel's integration with prompt flow in Azure Machine Learning enables users to perform detailed evaluations and manage experiments effectively ⁴⁸. Microsoft AutoGen includes built-in tools for tracking, tracing, and debugging agent interactions and workflows, and it also supports OpenTelemetry for broader observability. Furthermore, its integration with MLflow on Databricks allows for detailed trace logging of agent activities ²².

The AWS Multi-Agent Orchestrator offers sophisticated monitoring and analysis capabilities right out of the box, including comprehensive logging of interactions, agent overlap analysis to optimize agent roles, and built-in storage options for monitoring data ²⁹. The Databricks Mosaic AI Agent Framework provides built-in AI-assisted evaluation tools and an intuitive user interface that facilitates continuous improvement. It also integrates with Lakehouse Monitoring for tracking the quality of both data and AI models. Additionally, the Mosaic AI Gateway offers features for monitoring and governing the usage of AI models and agents ³⁴.

4. Local Experimentation and Setup

Experimenting with agentic orchestration frameworks locally, without the immediate need for cloud resources, can be valuable for initial exploration and prototyping. Several methods and resources are readily available to facilitate this process.

4.1. Methods for Local Development

The use of virtual environments in Python is highly recommended for isolating project dependencies, ensuring that different projects do not interfere with each other ²⁷. Tools like `venv` or `conda` can be used to create and manage these isolated environments. Docker provides a powerful way to create containerized environments, which guarantee consistency across different development systems ⁵³. Pre-configured Docker images for popular frameworks like LangChain, CrewAI, and AutoGen are available on Docker Hub, which can significantly simplify the setup process by providing ready-to-use environments with all the necessary dependencies ⁵³. This eliminates potential issues arising from inconsistencies in the development environment. Containerization ensures that the application runs in a self-contained and reproducible environment, regardless of the host system's specific configuration.

For experimentation that involves LLMs, frameworks such as Ollama and Llamafile allow developers to run LLMs directly on their personal devices ⁵⁷. This enables initial testing and development without the immediate reliance on paid cloud-based LLM APIs. LangChain, CrewAI, and AutoGen can all be configured to utilize these locally running LLMs. This approach offers a cost-effective and privacy-preserving way to experiment with agentic frameworks, particularly for tasks that do not require the most advanced or up-to-date models. Running LLMs locally reduces the dependency on external services and allows for experimentation even in environments with limited or no internet connectivity. LangGraph also offers a lightweight development server through its command-line interface (CLI), which supports hot reloading and debugging, making it suitable for local testing without the overhead of Docker ⁶⁰. Finally, Microsoft AutoGen provides AutoGen Studio, a no-code graphical user interface that allows users to rapidly prototype AI agents and workflows locally, further simplifying the initial experimentation phase ³.

4.2. Available Resources (Links and Setup Guides)

For those interested in getting hands-on experience with these frameworks locally, a wealth of resources is available:

LangChain:

- **Docker Image:** <https://hub.docker.com/r/langchain/langchain> ⁵³
- **Local Setup:** https://python.langchain.com/docs/how_to/local_llms/ ⁵⁷, <https://langchain-ai.github.io/langgraph/tutorials/langgraph-platform/local-server/> ⁶⁰
- **Tutorials:** https://python.langchain.com/docs/get_started/introduction
- **LangGraph Studio (Local Server):** <https://langchain-ai.github.io/langgraph/tutorials/langgraph-platform/local-server/> ⁶⁰, <https://langchain-ai.github.io/langgraph-studio/> ⁶⁵

LangGraph:

- **Local Setup (via LangChain CLI):** <https://langchain-ai.github.io/langgraph/tutorials/langgraph-platform/local-server/> ⁶⁰
- **GitHub Repository:** <https://github.com/langchain-ai/langgraph>

CrewAI:

- **Docker Image:** <https://hub.docker.com/r/tham0nk/crewai-studio> ⁵⁶, <https://hub.docker.com/r/crewai/crewai> ⁵⁴
- **Local Setup:** <https://docs.crewai.com/installation> ⁶⁶, <https://docs.crewai.com/quickstart> ⁶⁷
- **Tutorials:** <https://docs.crewai.com/how-to/first-crew>

Microsoft Semantic Kernel:

- **Local Setup (Python):** <https://learn.microsoft.com/en-us/semantic-kernel/get-started/quickstart>
- **GitHub Repository:** <https://github.com/microsoft/semantic-kernel>

Microsoft AutoGen:

- **Docker Image:** <https://github.com/microsoft/autogen/tree/main/.devcontainer> ⁵⁵
- **Local Setup:** <https://microsoft.github.io/autogen/docs/installation/> ⁶³, <https://microsoft.github.io/autogen/docs/autogen-studio/getting-started/> ⁶²
- **AutoGen Studio:** Instructions are included in the local setup guide.
- **GitHub Repository:** <https://github.com/microsoft/autogen>

AWS Multi-Agent Orchestrator:

- **Local Setup:** (<https://github.com/aws-labs/multi-agent-orchestrator/blob/main/README.md#getting-started>) ²⁸
- **GitHub Repository:** <https://github.com/aws-labs/multi-agent-orchestrator>

Databricks Mosaic AI Agent Framework: This framework primarily operates within a Databricks workspace. However, local development of tools and agents can be facilitated using the AI Bridge. Refer to the official Databricks documentation for detailed setup instructions within their environment.

5. Recommendations and Considerations

When selecting an agentic orchestration framework, several criteria should be carefully considered to align with specific project requirements and organizational capabilities.

5.1. Framework Selection Criteria

The scalability requirements of the intended agentic workflows are a primary consideration. Organizations anticipating a high volume and complexity of agent interactions should lean towards frameworks like AutoGen and LangGraph, which are designed to handle highly scalable systems. The ease of development is another crucial factor. For projects requiring rapid prototyping and addressing simpler use cases, CrewAI might offer a more accessible starting point. LangChain's comprehensive documentation also makes it a viable option for developers with varying levels of experience. The specific integration needs with AWS and Databricks services should be thoroughly evaluated. LangChain boasts the broadest range of integrations for both platforms, making it a strong contender for cross-platform solutions. The AWS Multi-Agent Orchestrator and Databricks Mosaic AI Agent Framework excel in providing deep integration within their respective cloud ecosystems. The level of monitoring and observability required for the agentic system should also be determined. The AWS Multi-Agent Orchestrator and Databricks Mosaic AI Agent Framework offer built-in monitoring tools, while LangChain and AutoGen integrate with dedicated monitoring platforms like LangSmith and MLflow. Finally, the existing infrastructure and the team's expertise should be taken into account, considering familiarity with programming languages (e.g., .NET for Semantic Kernel, Python for most others) and existing cloud infrastructure investments.

5.2. Top 3-5 Framework Recommendations

Based on the user's requirement for frameworks suitable for both AWS and Databricks environments, the following are the top recommendations:

1. **LangChain/LangGraph:** This combination offers the most comprehensive integration capabilities with both AWS and Databricks, supported by a large and active community. LangGraph, as an extension, provides advanced orchestration features for more complex multi-agent scenarios.
2. **CrewAI:** CrewAI provides a simpler and more intuitive approach to building multi-agent collaborations. It integrates effectively with both platforms through its reliance on LiteLLM for LLM connections and its ability to leverage LangChain tools.
3. **Microsoft AutoGen:** AutoGen is a strong choice for organizations looking to build scalable and highly automated multi-agent systems. It offers good integration with both AWS Bedrock and Databricks Foundation Model APIs and Unity Catalog.
4. **(Optional) Microsoft Semantic Kernel:** If the organization has a significant investment in the Microsoft ecosystem or prioritizes semantic reasoning in its AI applications, Semantic Kernel is a viable option. Its support for AWS is also steadily growing, making it increasingly relevant for multi-cloud scenarios.

5.3. Considerations for AWS and Databricks Specific Use Cases

For organizations primarily focused on the AWS ecosystem, the **AWS Multi-Agent Orchestrator** presents a compelling option due to its deep and native integration with various

AWS services. Similarly, for organizations that heavily leverage the Databricks platform for their data and AI initiatives, the **Databricks Mosaic AI Agent Framework** provides a tightly integrated and well-governed environment for building and deploying agentic applications within the Databricks ecosystem.

5.4. Comparative Table of Recommended Frameworks

The following table provides a comparative overview of the key features of the top recommended frameworks:

Feature	LangChain/LangGraph	CrewAI	Microsoft AutoGen	Microsoft Semantic Kernel
Scalability	Good (modular), Excellent (LangGraph's graph)	Good (for team-based tasks)	Excellent (asynchronous, event-driven)	Good (enterprise-focused, asynchronous)
Ease of Use	Moderate (versatile, but tuning needed)	Easy (intuitive, rapid prototyping)	Moderate (no-code Studio, but setup can be involved)	Moderate (combines AI with programming)
Integration (AWS)	Extensive (Bedrock, Kendra, SageMaker, etc.)	Good (via LiteLLM for Bedrock, SageMaker)	Good (Bedrock integration)	Good (direct Bedrock Agents and Bedrock support)
Integration (DB)	Extensive (SQL, HuggingFace, Databricks models)	Good (via LiteLLM, Unity Catalog)	Good (DBRX, Unity Catalog)	Moderate (Python support allows potential integration)

Monitoring	LangSmith, MLflow	CloudWatch (AWS), MLflow (Databricks)	Built-in tools, OpenTelemetry , MLflow	Prompt flow (Azure ML)
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6. Conclusion: Summary of Findings and Future Outlook for Agentic Orchestration

The landscape of agentic orchestration frameworks offers a diverse set of tools catering to various needs and environments. LangChain and LangGraph stand out for their extensive integration capabilities across both AWS and Databricks, providing a solid foundation for a wide range of applications. CrewAI offers a more streamlined approach focused on multi-agent collaboration, making it ideal for rapid prototyping and team-based tasks. Microsoft AutoGen excels in building highly scalable and automated systems, with growing support for both cloud platforms. Microsoft Semantic Kernel is a strong contender for organizations within the Microsoft ecosystem or those prioritizing semantic reasoning, with increasing integration with AWS. While the AWS Multi-Agent Orchestrator and Databricks Mosaic AI Agent Framework offer deep integration within their respective cloud environments, LangChain, CrewAI, and AutoGen provide more versatile options for organizations operating across both AWS and Databricks.

The field of agentic AI is characterized by rapid evolution, with new frameworks, features, and integrations emerging frequently²⁷. This dynamic nature suggests that the information presented herein reflects a current state, and continuous evaluation of the landscape will be necessary for organizations to make informed decisions. The increasing importance of agentic AI in driving automation, enhancing decision-making, and fostering innovation indicates a promising future for these frameworks. As LLM technology continues to advance and cloud platforms expand their AI service offerings, agentic orchestration frameworks will likely become even more sophisticated, user-friendly, and tightly integrated with the underlying infrastructure, further empowering developers to build intelligent and autonomous systems at scale.

Works cited

1. Agentic Systems and Databricks Genie: The Future of Autonomous Data Processing, accessed on March 18, 2025, <https://sutejakanuri.medium.com/agentic-systems-and-databricks-genie-the-future-of-autonomous-data-processing-1c083fde298d>
2. Top 5 Agentic AI Frameworks to Watch in 2025 - Cloudester, accessed on March 18, 2025, <https://cloudester.com/top-agentic-ai-frameworks-to-watch/>
3. Top 5 Agentic AI Frameworks You Should Know in 2025 - Hyperstack, accessed on March 18, 2025, <https://www.hyperstack.cloud/blog/case-study/top-agentic-ai-frameworks-you-should-know>
4. Top 5 Agentic AI Frameworks to Watch in 2025 | by Lekha Priya - Medium, accessed on March 18, 2025, <https://lekha-bhan88.medium.com/top-5-agentic-ai-frameworks-to-watch-in-2025-9d51b2b652c0>
5. Agentic Frameworks: A Guide to the Systems Used to Build AI Agents - Moveworks, accessed on March 18, 2025, <https://www.moveworks.com/us/en/resources/blog/what-is-agentic-framework>
6. AI Agent Frameworks: Choosing the Right Foundation for Your Business | IBM, accessed on

March 18, 2025, <https://www.ibm.com/think/insights/top-ai-agent-frameworks>

7. What is LangChain? - AWS, accessed on March 18, 2025,

<https://aws.amazon.com/what-is/langchain/>

8. AWS - LangChain, accessed on March 18, 2025,

<https://python.langchain.com/docs/integrations/providers/aws/>

9. Databricks Tutorial For LangChain | Restackio, accessed on March 18, 2025,

<https://www.restack.io/p/natural-language-processing-databricks-answer-cat-ai>

10. How to Integrate Databricks LLM with Langchain Efficiently | by Gary Svenson | Towards AGI, accessed on March 18, 2025,

<https://medium.com/towards-agi/how-to-integrate-databricks-llm-with-langchain-efficiently-56df79541211>

11. LangChain on Azure Databricks for LLM development - Microsoft Learn, accessed on March 18, 2025, <https://learn.microsoft.com/en-us/azure/databricks/large-language-models/langchain>

12. Build a Multi-Agent System with LangGraph and Mistral on AWS, accessed on March 18, 2025,

<https://aws.amazon.com/blogs/machine-learning/build-a-multi-agent-system-with-langgraph-and-mistral-on-aws/>

13. Author AI agents in code - Databricks Documentation, accessed on March 18, 2025,

<https://docs.databricks.com/aws/en/generative-ai/agent-framework/author-agent>

14. Get started with Databricks AI agents - Microsoft Learn, accessed on March 18, 2025,

<https://learn.microsoft.com/en-us/azure/databricks/generative-ai/tutorials/agent-framework-notebook>

15. Get started with Databricks AI agents, accessed on March 18, 2025,

<https://docs.databricks.com/aws/en/generative-ai/tutorials/agent-framework-notebook>

16. Build agentic AI solutions with DeepSeek-R1, CrewAI, and Amazon SageMaker AI - AWS, accessed on March 18, 2025,

<https://aws.amazon.com/blogs/machine-learning/build-agentic-ai-solutions-with-deepseek-r1-crewai-and-amazon-sagemaker-ai/>

17. Connect to any LLM - CrewAI, accessed on March 18, 2025,

<https://docs.crewai.com/how-to/llm-connections>

18. CrewAI new features. Agentic AI framework | by Xin Cheng | Feb, 2025 - Medium, accessed on March 18, 2025, <https://billtcheng2013.medium.com/crewai-new-features-ff79f55cdc79>

19. CrewAI - Unity Catalog, accessed on March 18, 2025,

<https://docs.unitycatalog.io/ai/integrations/crewai/>

20. Integration of AWS Bedrock Agents in Semantic Kernel - Microsoft Developer Blogs, accessed on March 18, 2025,

<https://devblogs.microsoft.com/semantic-kernel/integration-of-aws-bedrock-agents-in-semantic-kernel/>

21. Introducing AWS Bedrock with Semantic Kernel - Microsoft Developer Blogs, accessed on March 18, 2025,

<https://devblogs.microsoft.com/semantic-kernel/introducing-aws-bedrock-with-semantic-kernel/>

22. AutoGen - Microsoft Research, accessed on March 18, 2025,

<https://www.microsoft.com/en-us/research/project/autogen/>

23. Amazon Bedrock | AutoGen 0.2 - Microsoft Open Source, accessed on March 18, 2025,

<https://microsoft.github.io/autogen/0.2/docs/topics/non-openai-models/cloud-bedrock/>

24. Add support for AWS Bedrock integration · Issue #5226 · microsoft/autogen - GitHub, accessed on March 18, 2025, <https://github.com/microsoft/autogen/issues/5226>

25. AutoGen - Unity Catalog, accessed on March 18, 2025,

<https://docs.unitycatalog.io/ai/integrations/autogen/>

26. Use AutoGen in Databricks with DBRX - Microsoft Open Source, accessed on March 18, 2025, https://microsoft.github.io/autogen/0.2/docs/notebooks/agentchat_databricks_dbrx/

27. AWS Multi-Agent Orchestrator: A Guide With Examples - DataCamp, accessed on March 18, 2025, <https://www.datacamp.com/tutorial/aws-multi-agent-orchestrator>

28. awslabs/multi-agent-orchestrator: Flexible and powerful framework for managing multiple AI agents and handling complex conversations - GitHub, accessed on March 18, 2025, <https://github.com/awslabs/multi-agent-orchestrator>

29. Introducing AWS Multi-Agent Orchestrator: The Symphony of AI Agents - Medium, accessed on March 18, 2025, <https://medium.com/@cerizzi/introducing-aws-multi-agent-orchestrator-the-symphony-of-ai-agents-41a6a87de927>

30. AWS Introduces Multi-Agent Orchestrator Framework, accessed on March 18, 2025, <https://www.aiagentpulse.com/p/aws-introduces-multi-agent-orchestrator-framework>

31. AWS Reveals Multi-Agent Orchestrator Framework for Managing AI Agents - InfoQ, accessed on March 18, 2025, <https://www.infoq.com/news/2024/12/aws-multi-agent/>

32. Agentic AI on AWS, the design patterns and considerations, accessed on March 18, 2025, <https://community.aws/content/2qlet7NAUR02ygnurN1O74ClwTN/agentic-ai-framework-on-aws-the-design-patterns-and-considerations>

33. Design multi-agent orchestration with reasoning using Amazon Bedrock and open source frameworks | AWS Machine Learning Blog, accessed on March 18, 2025, <https://aws.amazon.com/blogs/machine-learning/design-multi-agent-orchestration-with-reasoning-using-amazon-bedrock-and-open-source-frameworks/>

34. Mosaic AI Agent Framework - Databricks, accessed on March 18, 2025, <https://www.databricks.com/product/machine-learning/retrieval-augmented-generation>

35. Agentic RAG Framework and Databricks Mosaic AI | by Sandip Roy - Medium, accessed on March 18, 2025, <https://roysandip.medium.com/agentic-rag-framework-and-databricks-mosaic-ai-e037c8323a51>

36. Build an Autonomous AI Assistant with Mosaic AI Agent Framework | Databricks Blog, accessed on March 18, 2025, <https://www.databricks.com/blog/build-autonomous-ai-assistant-mosaic-ai-agent-framework>

37. Build Compound AI Systems Faster with Databricks Mosaic AI, accessed on March 18, 2025, <https://www.databricks.com/blog/build-compound-ai-systems-faster-databricks-mosaic-ai>

38. Architecture for building an GenAI application with Databricks Mosaic AI and Unity Catalog | by Pawan Kumar Shukla | Feb, 2025 | Medium, accessed on March 18, 2025, https://medium.com/@pawankumarshukla_57258/building-an-enterprise-ai-chatbot-with-databricks-mosaic-ai-and-unity-catalog-f959a89ebfc4

39. 3 Databricks Mosaic AI Use Cases to Supercharge Your Log Analytics Program, accessed on March 18, 2025, <https://www.chaossearch.io/blog/mosaic-ai-use-cases>

40. Create an AI agent - Azure Databricks | Microsoft Learn, accessed on March 18, 2025, <https://learn.microsoft.com/en-us/azure/databricks/generative-ai/agent-framework/create-agent>

41. Building a RAG System with Amazon Bedrock and Databricks - Medium, accessed on March 18, 2025, <https://medium.com/@AI-on-Databricks/building-a-rag-system-with-amazon-bedrock-and-databricks-a7cc47bcb2ff>

42. Zero to generative AI with Databricks and AWS | AWS Partner Network (APN) Blog, accessed on March 18, 2025, <https://aws.amazon.com/blogs/apn/zero-to-generative-ai-with-databricks-and-aws/>

43. Agents: Mosaic AI, Tools, and Function Calling | by AI on Databricks | Jan, 2025 - Medium, accessed on March 18, 2025, <https://medium.com/@AI-on-Databricks/agents-mosaic-ai-tools-and-function-calling-2530c841f80c>
44. Revolutionizing AWS Security with AI Agents and CrewAI - Paul M. Duvall, accessed on March 18, 2025, <https://www.paulmduvall.com/ai-agents-for-aws-security-businesses/>
45. AI Integrations for Semantic Kernel - Microsoft Learn, accessed on March 18, 2025, <https://learn.microsoft.com/en-us/semantic-kernel/concepts/ai-services/integrations>
46. Databricks Autologging - Azure Databricks | Microsoft Learn, accessed on March 18, 2025, <https://learn.microsoft.com/en-us/azure/databricks/mlflow/databricks-autologging>
47. Generative AI app developer workflow - Databricks Documentation, accessed on March 18, 2025, <https://docs.databricks.com/aws/en/generative-ai/tutorials/ai-cookbook/genai-developer-workflow>
48. Evaluate Semantic Kernel with prompt flow - Azure Machine Learning | Microsoft Learn, accessed on March 18, 2025, <https://learn.microsoft.com/en-us/azure/machine-learning/prompt-flow/how-to-evaluate-semantic-kernel?view=azureml-api-2>
49. MLflow for gen AI agent and ML model lifecycle - Azure Databricks - Microsoft Learn, accessed on March 18, 2025, <https://learn.microsoft.com/en-us/azure/databricks/mlflow/>
50. Build and deploy quality AI agent systems - Databricks, accessed on March 18, 2025, <https://www.databricks.com/product/artificial-intelligence>
51. How to Create a Free Langchain Application Step by Step | by Gary Svenson - Medium, accessed on March 18, 2025, <https://medium.com/@garysvenson09/how-to-create-a-free-langchain-application-step-by-step-7b04517a858f>
52. Developing with CrewAI - Patrick Rachford - Medium, accessed on March 18, 2025, <https://patford12.medium.com/developing-with-crewai-086d3aafe9de>
53. langchain/langchain - Docker Image - Docker Hub, accessed on March 18, 2025, <https://hub.docker.com/r/langchain/langchain>
54. Crewai docker integration guide | Restackio, accessed on March 18, 2025, <https://www.restack.io/p/crewai-answer-docker-integration-guide-cat-ai>
55. Docker | AutoGen 0.2, accessed on March 18, 2025, <https://microsoft.github.io/autogen/0.2/docs/installation/Docker/>
56. tham0nk/crewai-studio - Docker Image, accessed on March 18, 2025, <https://hub.docker.com/r/tham0nk/crewai-studio>
57. Run models locally | 🦜 LangChain, accessed on March 18, 2025, https://python.langchain.com/docs/how_to/local_llms/
58. Integrate CrewAI with Ollama Locally and Privately to Run AI Agents - YouTube, accessed on March 18, 2025, https://www.youtube.com/watch?v= 5_QHE3p8EA
59. Local AI Agents with CrewAI (and Ollama) - FOSS Engineer, accessed on March 18, 2025, <https://fossengineer.com/ai-agents-crewai/>
60. Quickstart: Launch Local LangGraph Server - GitHub Pages, accessed on March 18, 2025, <https://langchain-ai.github.io/langgraph/tutorials/langgraph-platform/local-server/>
61. LangGraph CLI - GitHub Pages, accessed on March 18, 2025, <https://langchain-ai.github.io/langgraph/cloud/reference/cli/>
62. AutoGen Studio - Getting Started, accessed on March 18, 2025, <https://microsoft.github.io/autogen/0.2/docs/autogen-studio/getting-started/>

63. microsoft/autogen: A programming framework for agentic AI PyPi: autogen-agentchat
Discord: <https://aka.ms/autogen-discord> Office Hour: <https://aka.ms/autogen-officehour> -
GitHub, accessed on March 18, 2025, <https://github.com/microsoft/autogen>
64. How to Install Microsoft AutoGen 0.4 with AutoGen Studio Locally? - NodeShift, accessed
on March 18, 2025,
<https://nodeshift.com/blog/how-to-install-microsoft-autogen-0-4-with-autogen-studio-locally>
65. LangGraph Studio Desktop (Beta) - GitHub, accessed on March 18, 2025,
<https://github.com/langchain-ai/langgraph-studio>
66. Installation - CrewAI, accessed on March 18, 2025, <https://docs.crewai.com/installation>
67. Quickstart - CrewAI, accessed on March 18, 2025, <https://docs.crewai.com/quickstart>